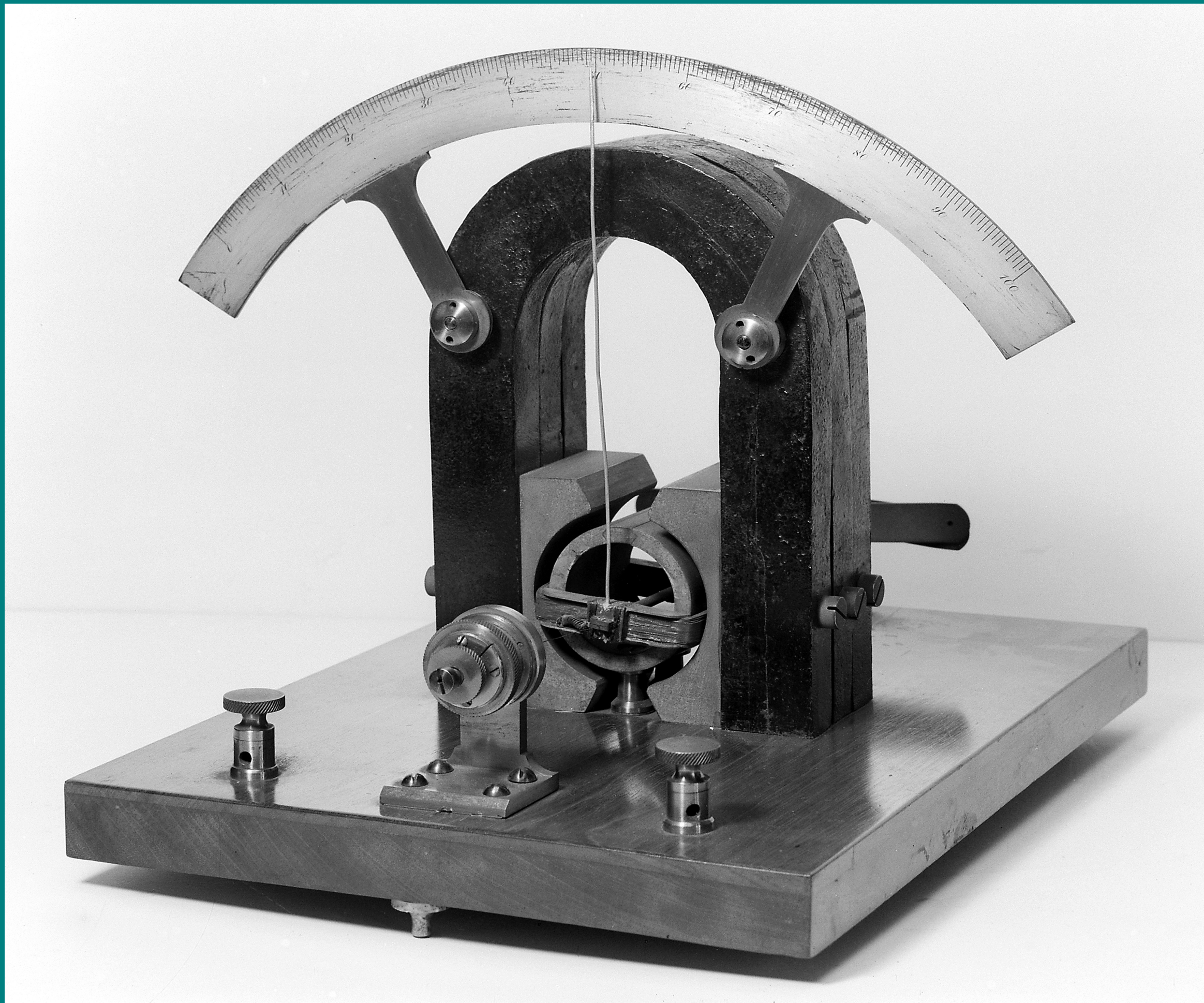


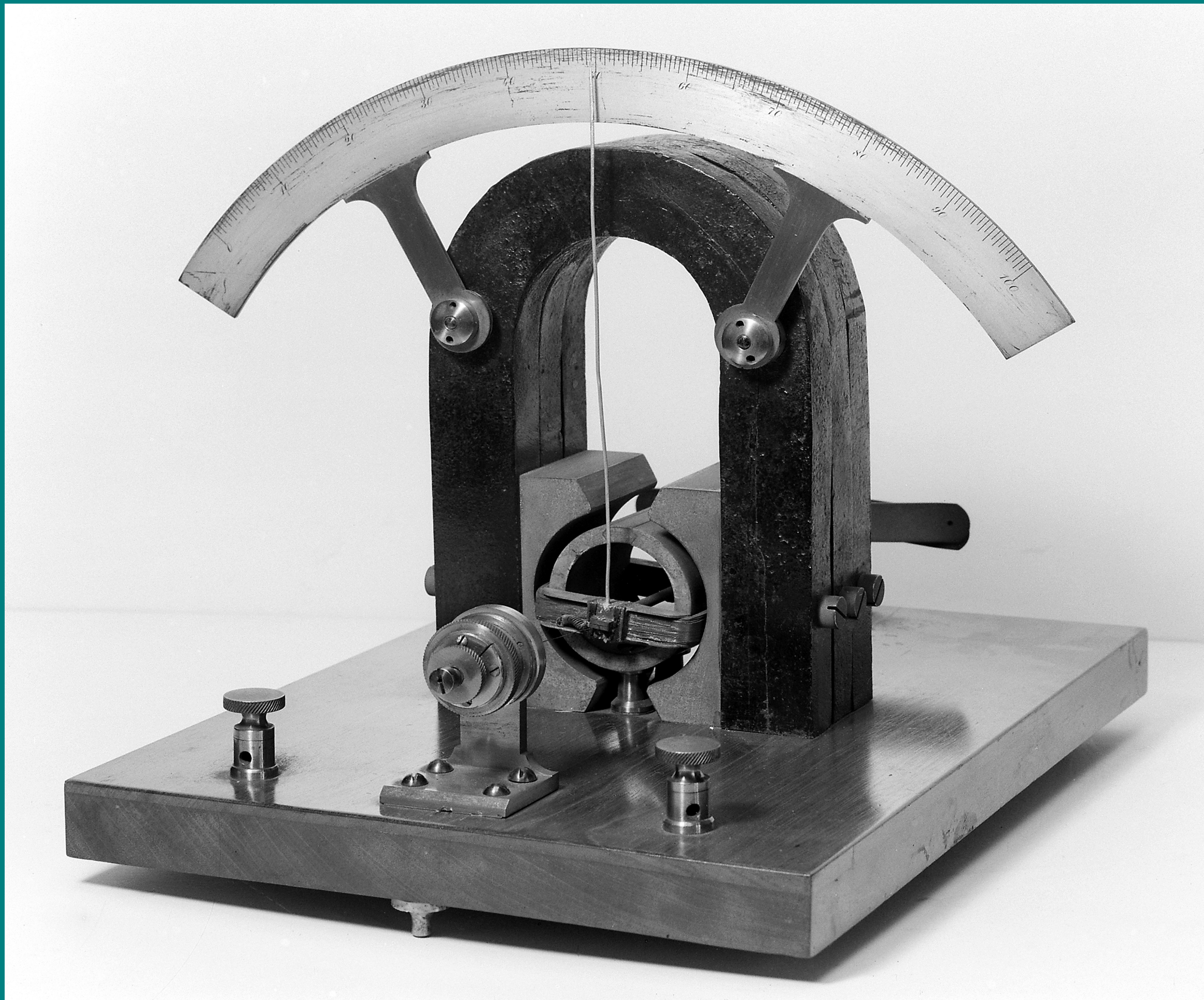
THE MOVING COIL GALVANOMETER

Moving Coil Galvanometer (MCG)

- A **moving coil galvanometer** is an instrument which is used to detect and measure electric currents.



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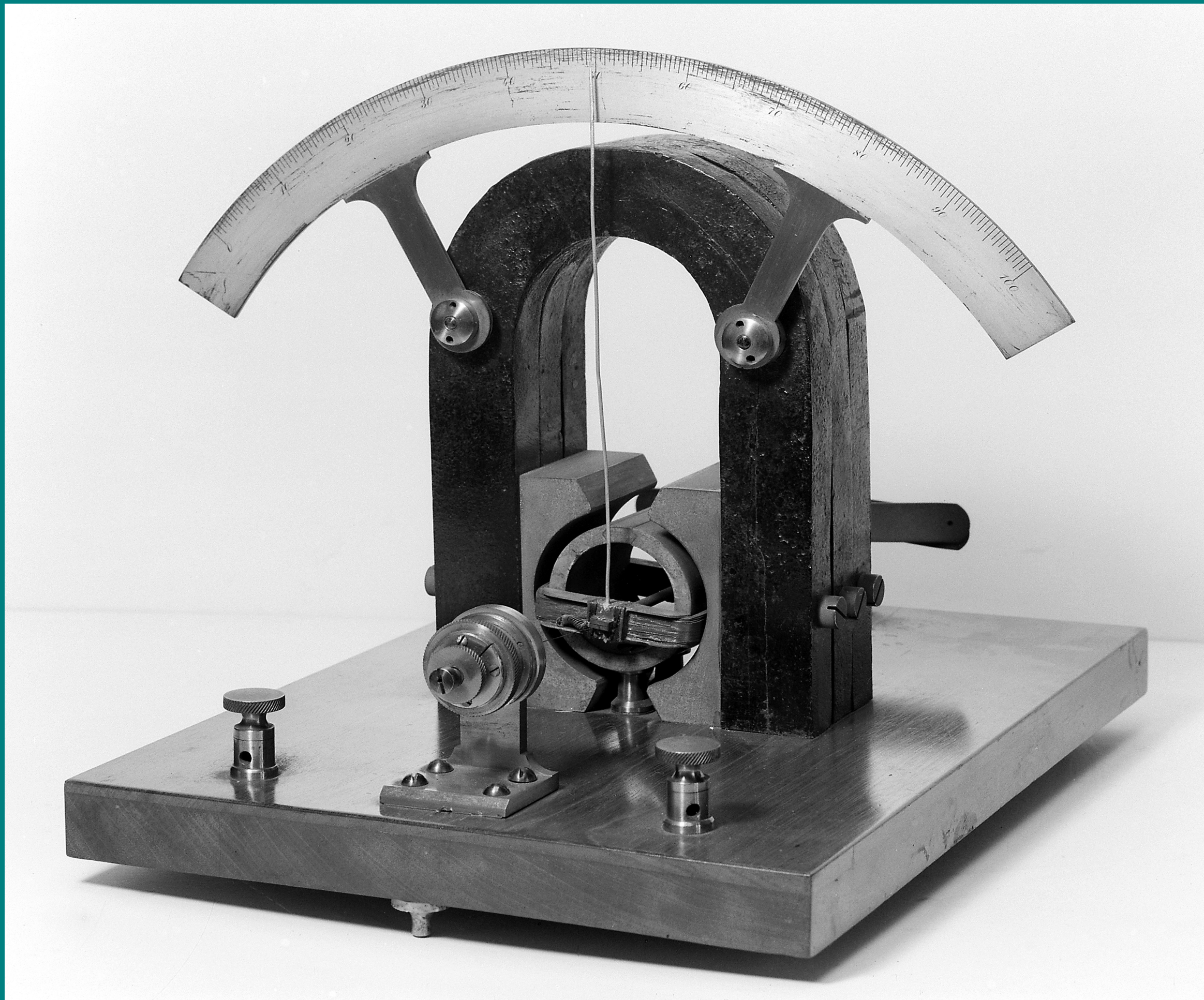


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Working Principle of A MCG

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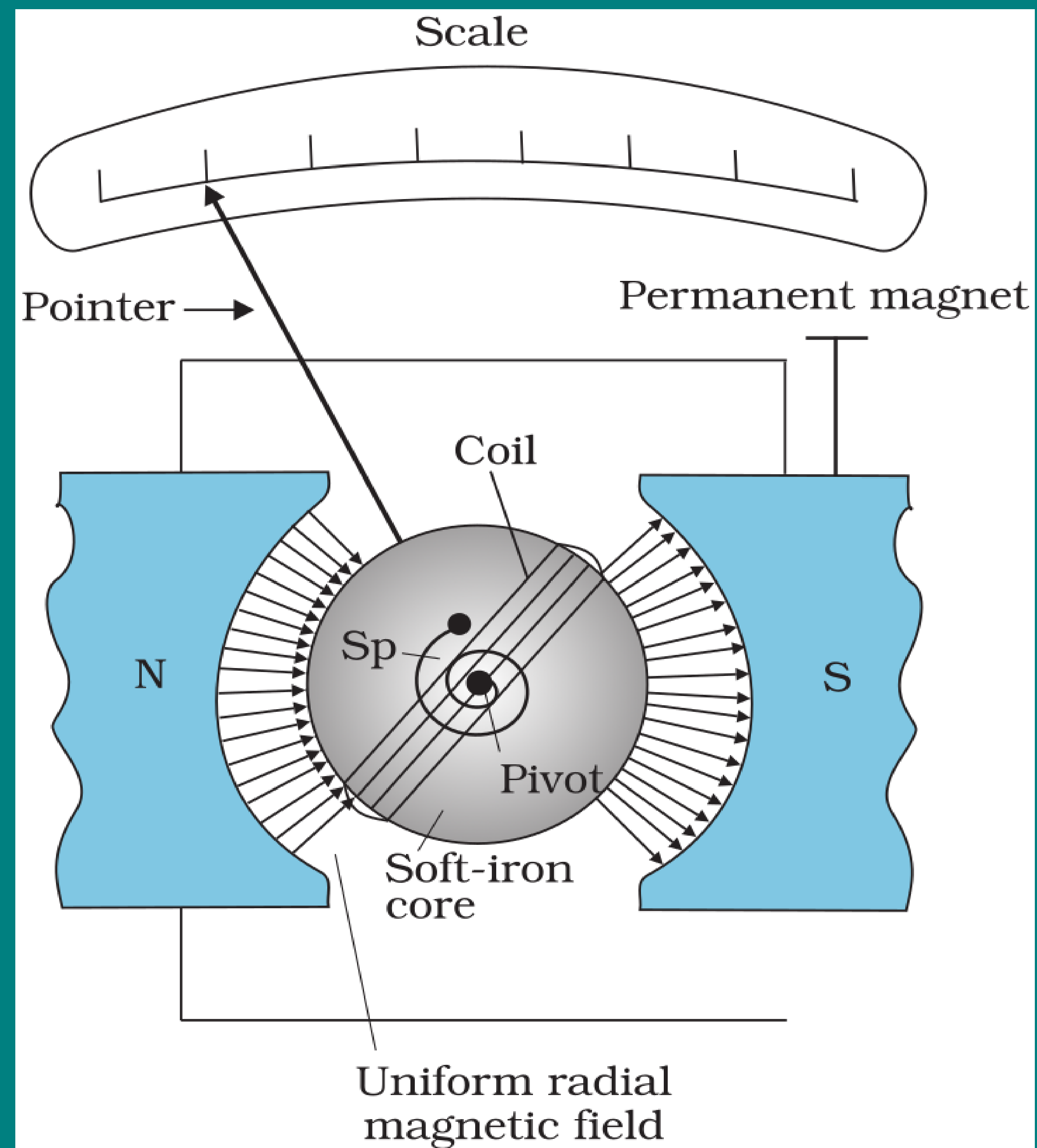
Moving Coil Galvanometer (MCG)

- A **moving coil galvanometer** is an instrument which is **used to detect and measure electric currents**.

Working Principle of A MCG

- A **current-carrying coil**, when **placed in an external magnetic field**, **experiences a torque**, which **deflects the coil**. and the **deflection is directly proportional to the amount of the current** passed through the coil.

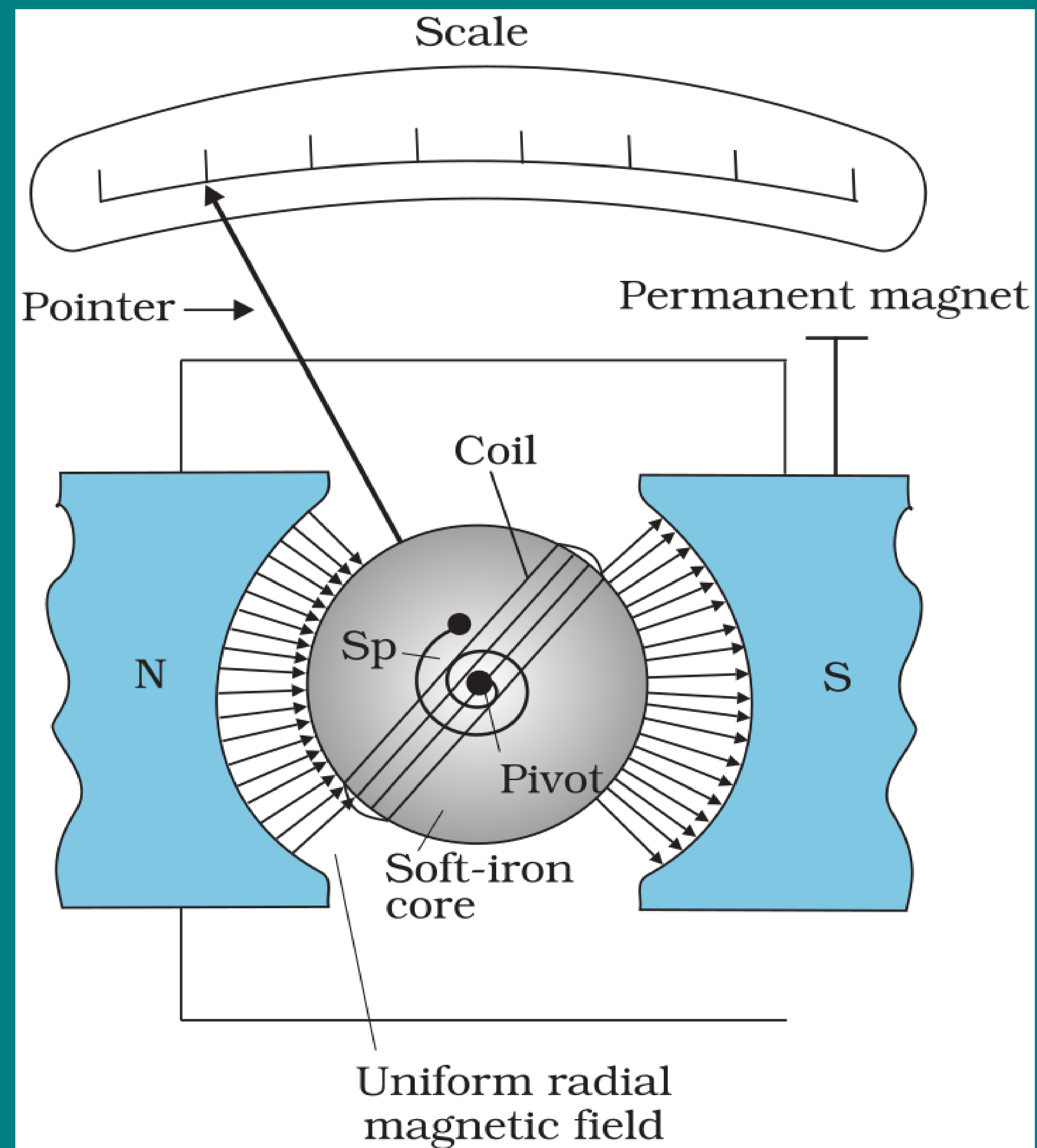
THE MOVING COIL GALVANOMETER



Construction of A (MCG)

- The **galvanometer consists of a coil**, with **many turns (N)**, **free to rotate** about a fixed axis in a **uniform radial magnetic field B**

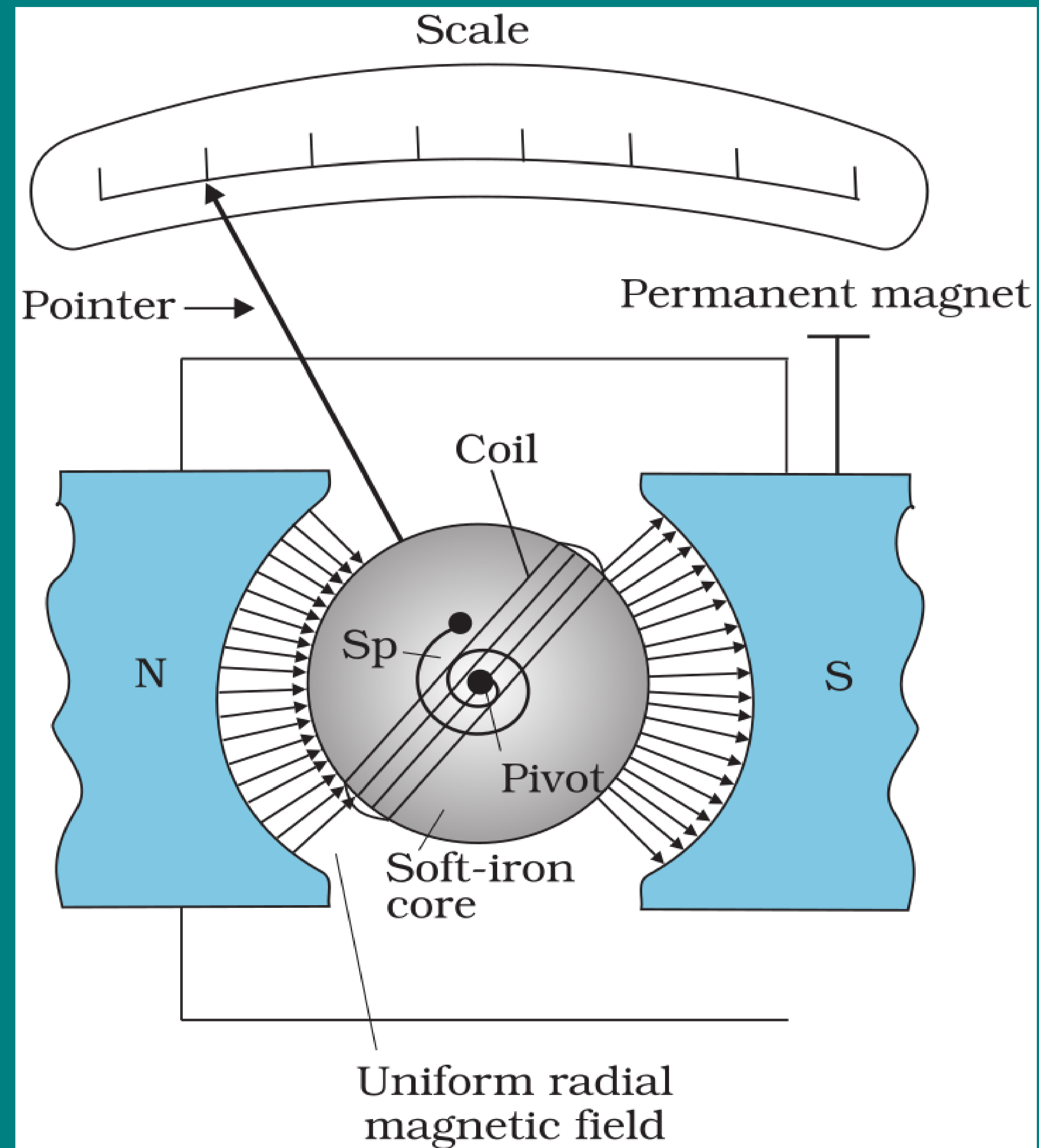
THE MOVING COIL GALVANOMETER



Construction of A (MCG)

- The **galvanometer consists of a coil**, with **many turns (N)**, free to rotate about a fixed axis in a **uniform radial magnetic field B**
- There is a **cylindrical soft iron core** which not only **makes the field radial** but also **increases the strength of the magnetic field**.

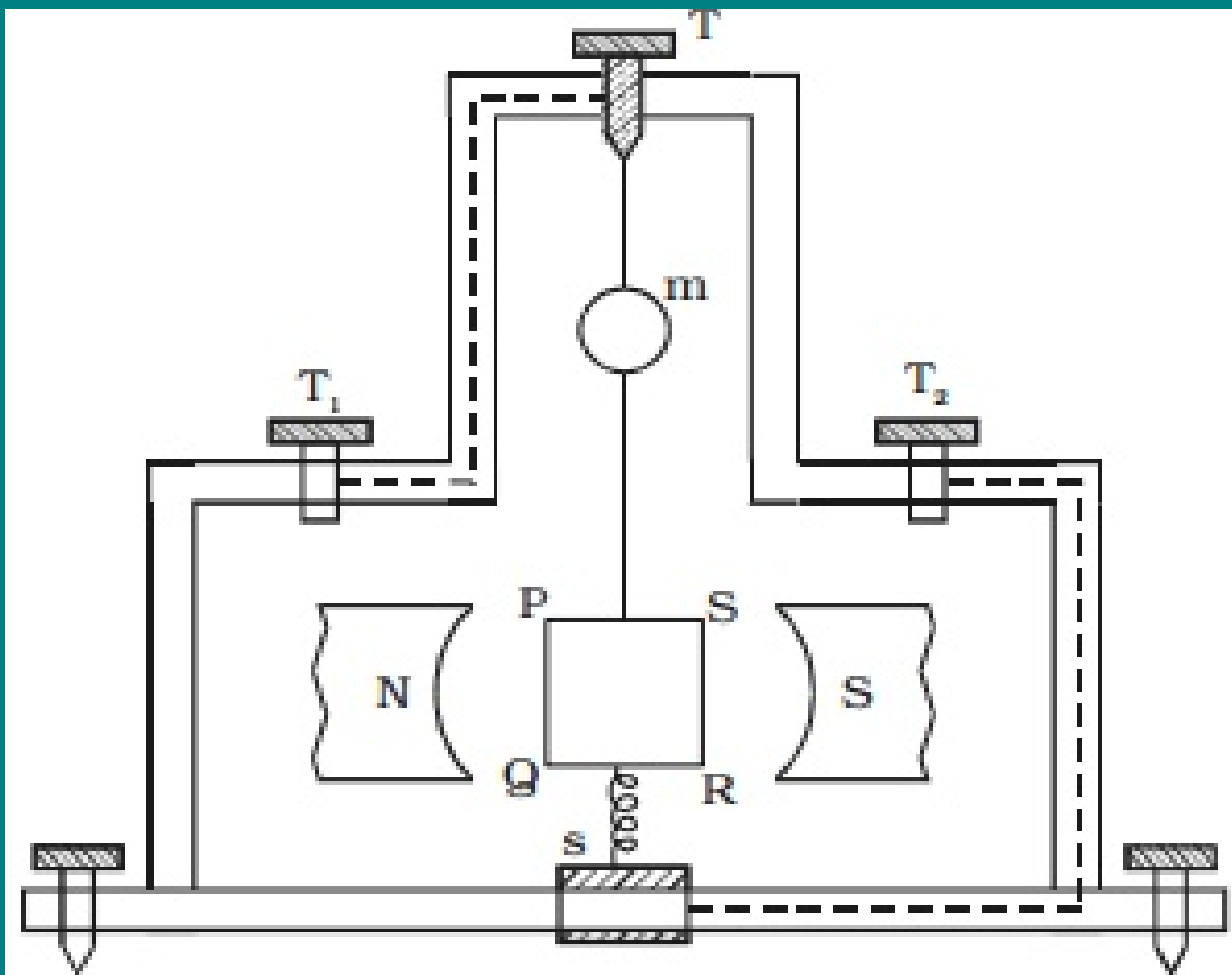
THE MOVING COIL GALVANOMETER



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- The **galvanometer consists of a coil**, with **many turns (N)**, free to rotate about a fixed axis in a **uniform radial magnetic field B**
- There is a **cylindrical soft iron core** which not only **makes the field radial** but also **increases the strength of the magnetic field**.
- The **lower end of the coil is connected to a spring (S_P)**

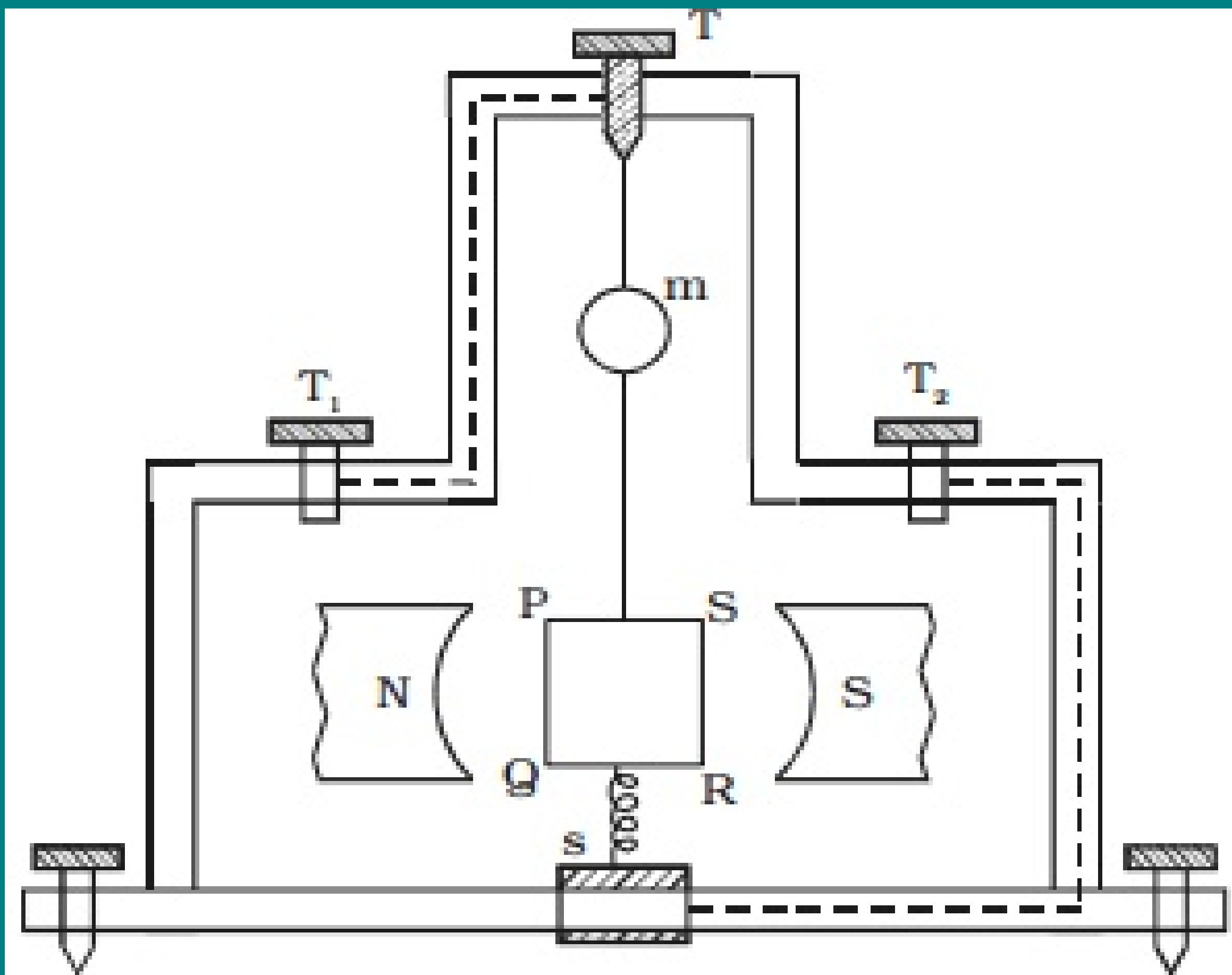
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- When a current (I) flows through the coil, a **torque** acts on it. This torque is given by

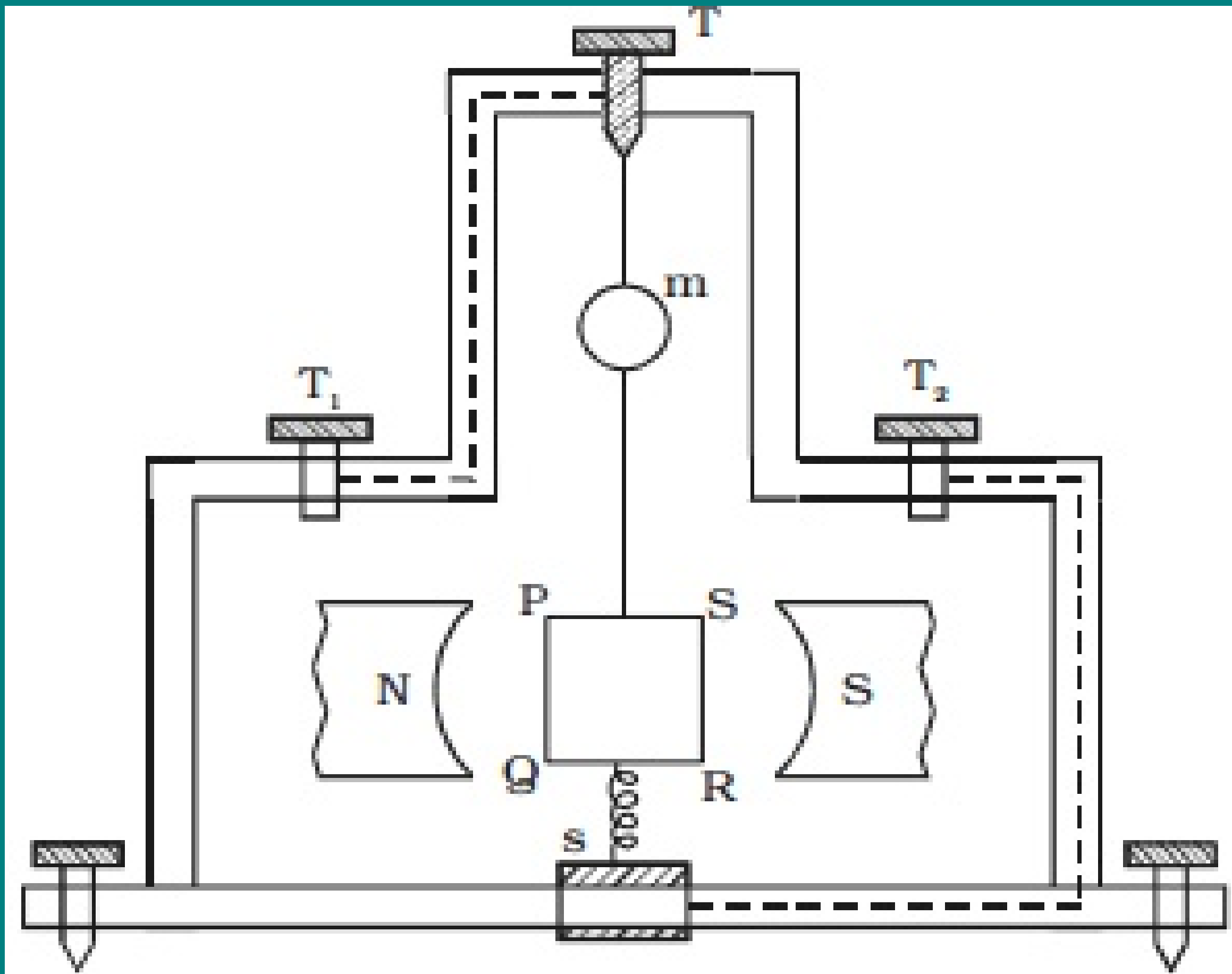
THE MOVING COIL GALVANOMETER



Working of A (MCG)

- **When a current (I) flows** through the coil, a **torque acts on it**. This torque is given by $\tau = N I A B$ ($\sin \theta = 1$ since the field is radial.)

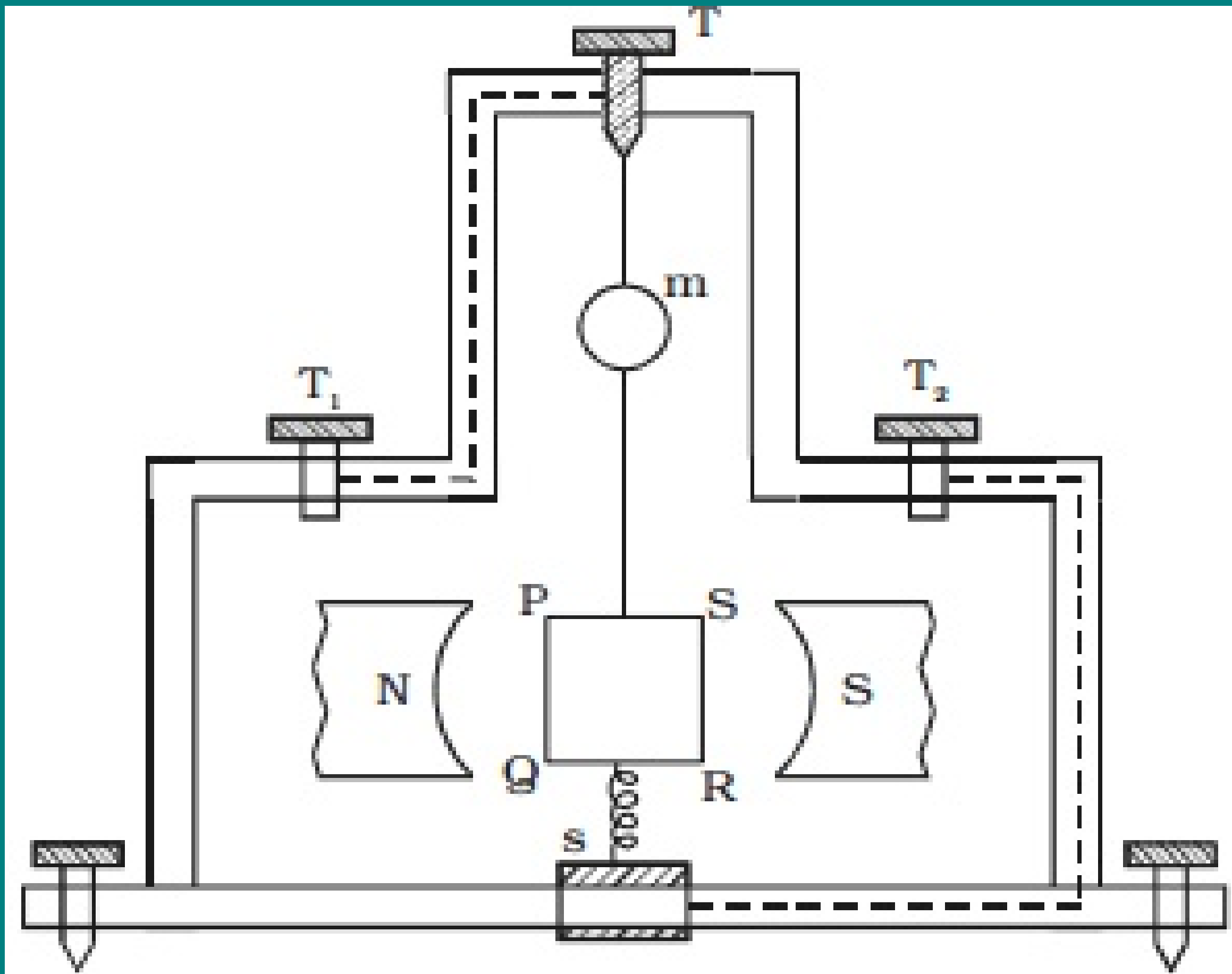
THE MOVING COIL GALVANOMETER



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- When a current (I) flows through the coil, a torque acts on it. This torque is given by $\tau = N I A B$ ($\sin \theta = 1$ since the field is radial.)
- This magnetic torque ($N I A B$) tends to rotate the coil.

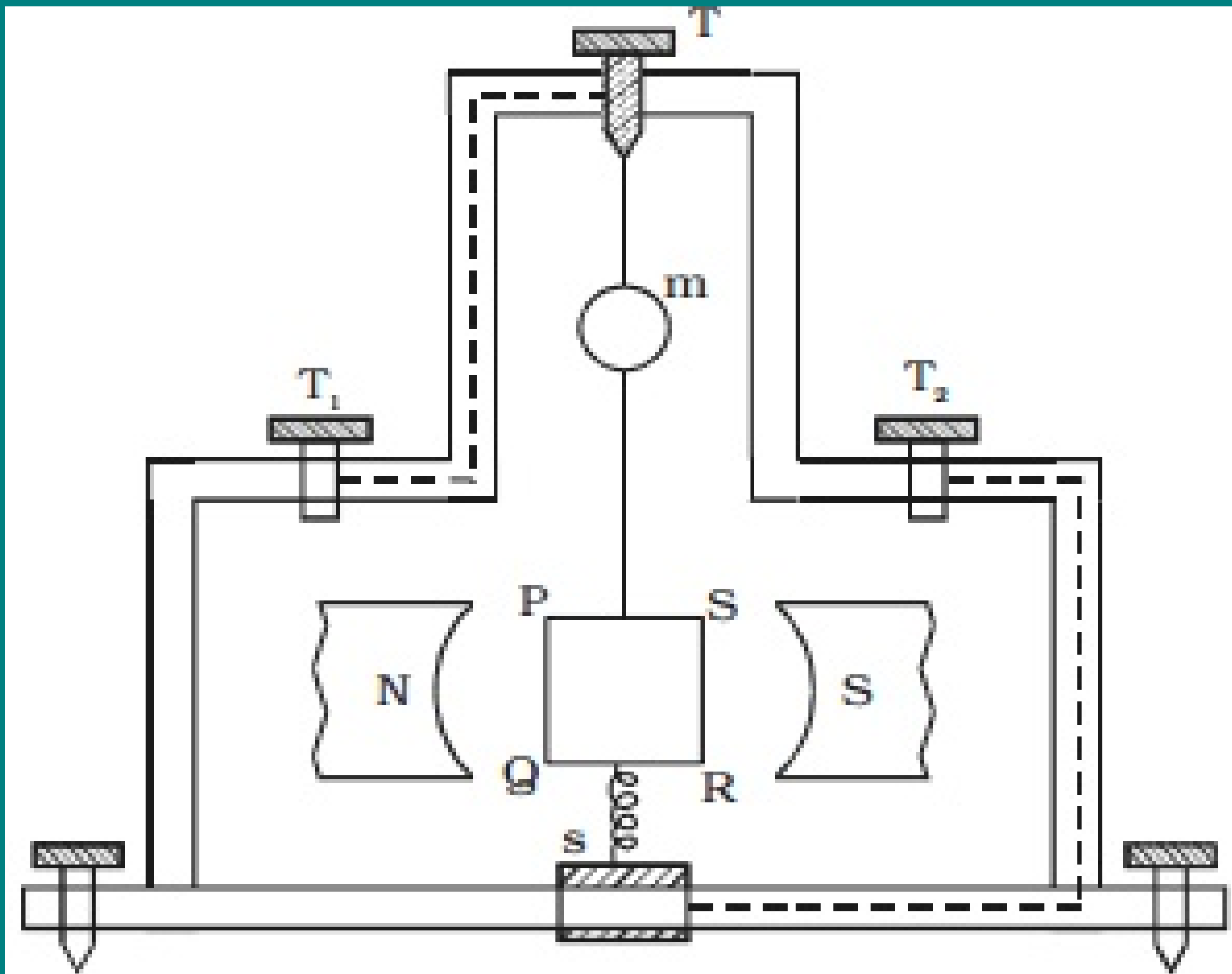
THE MOVING COIL GALVANOMETER



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- The spring provides a counter torque ($k\phi$) that balances the magnetic torque ($N I A B$); resulting in a steady angular deflection (ϕ).

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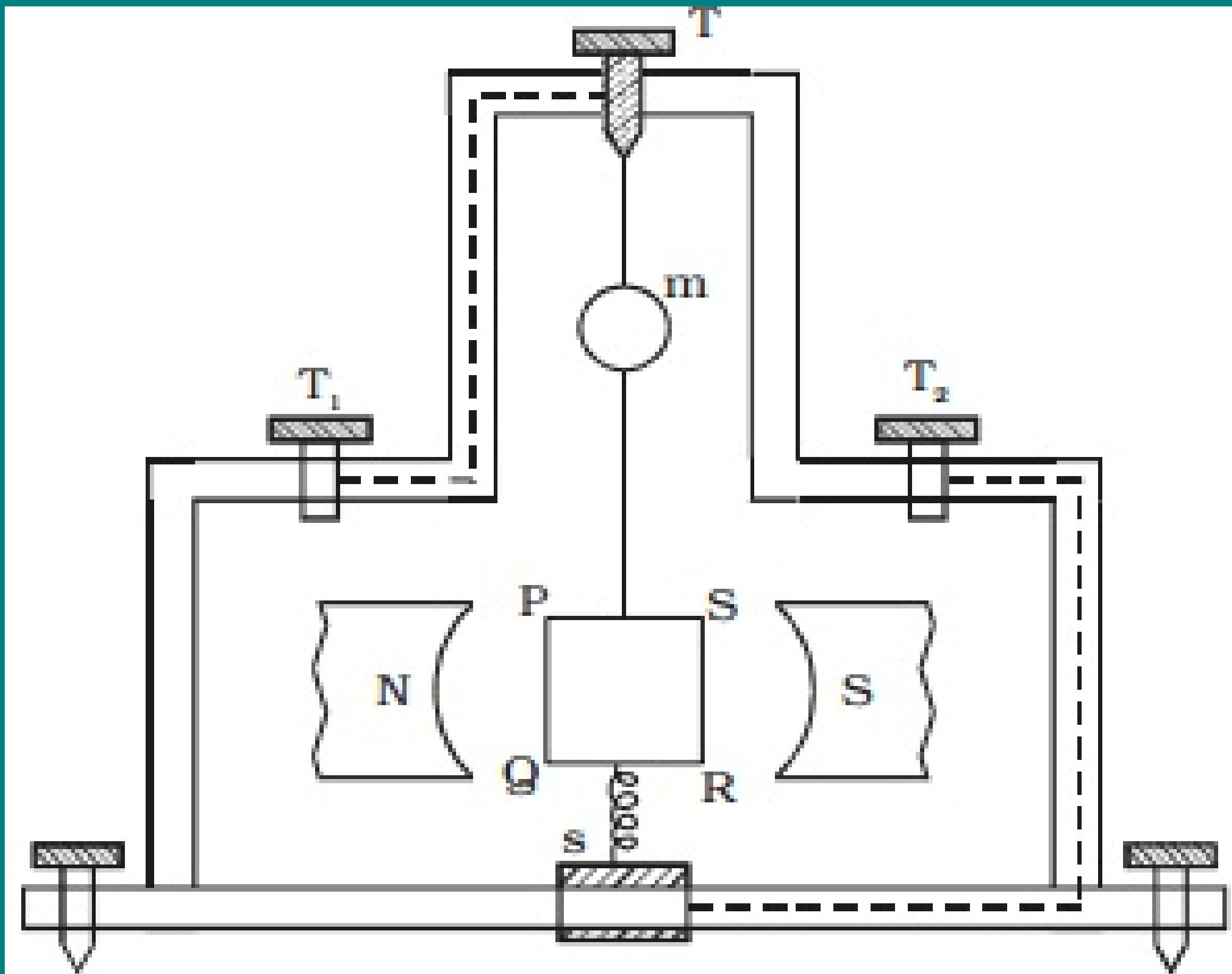


Working of A (MCG)

- In equilibrium

$$k\phi = N I A B$$

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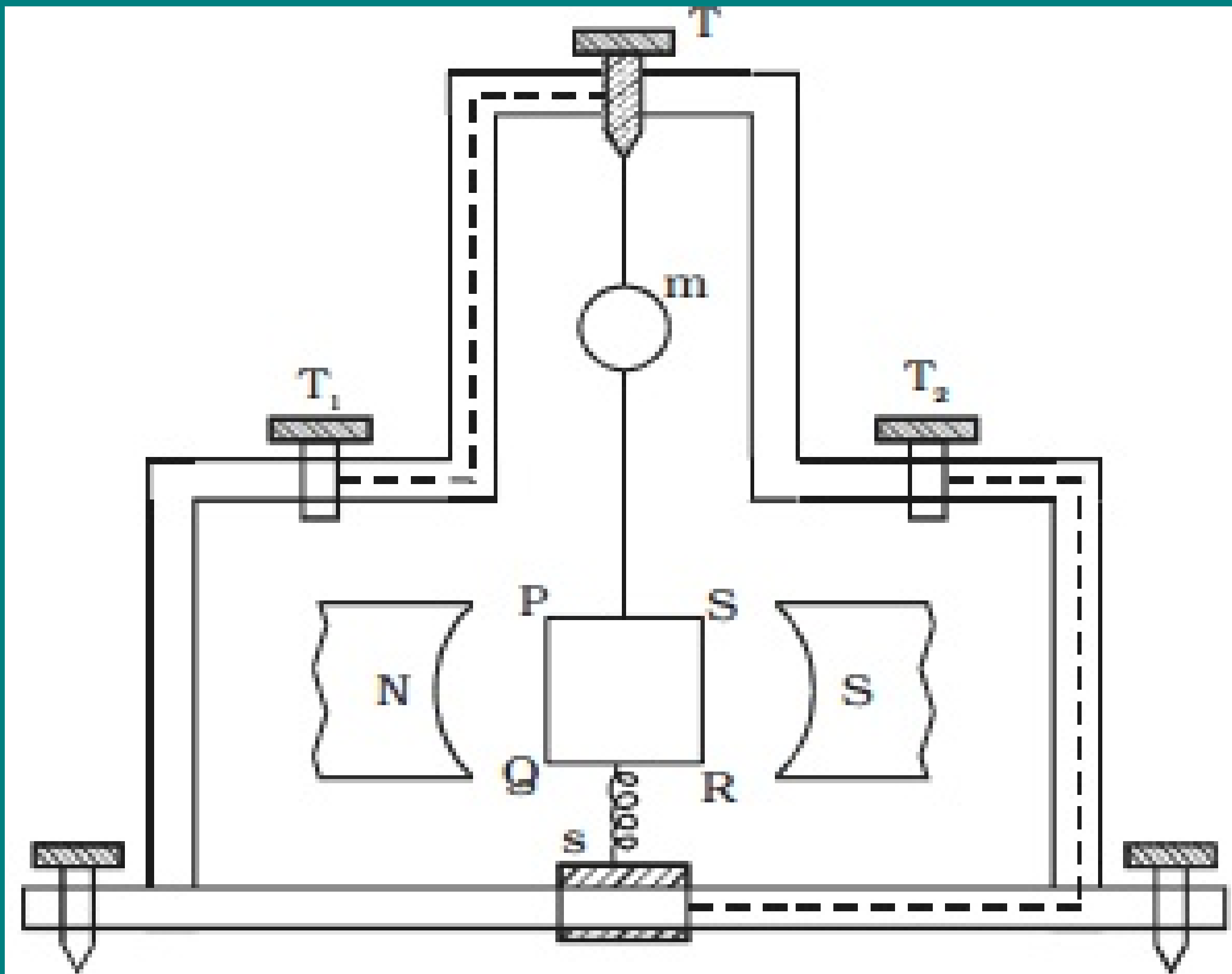


Working of A (MCG)

- In equilibrium

$$k\phi = NIAB$$
$$\phi = \frac{NIAB}{k}$$

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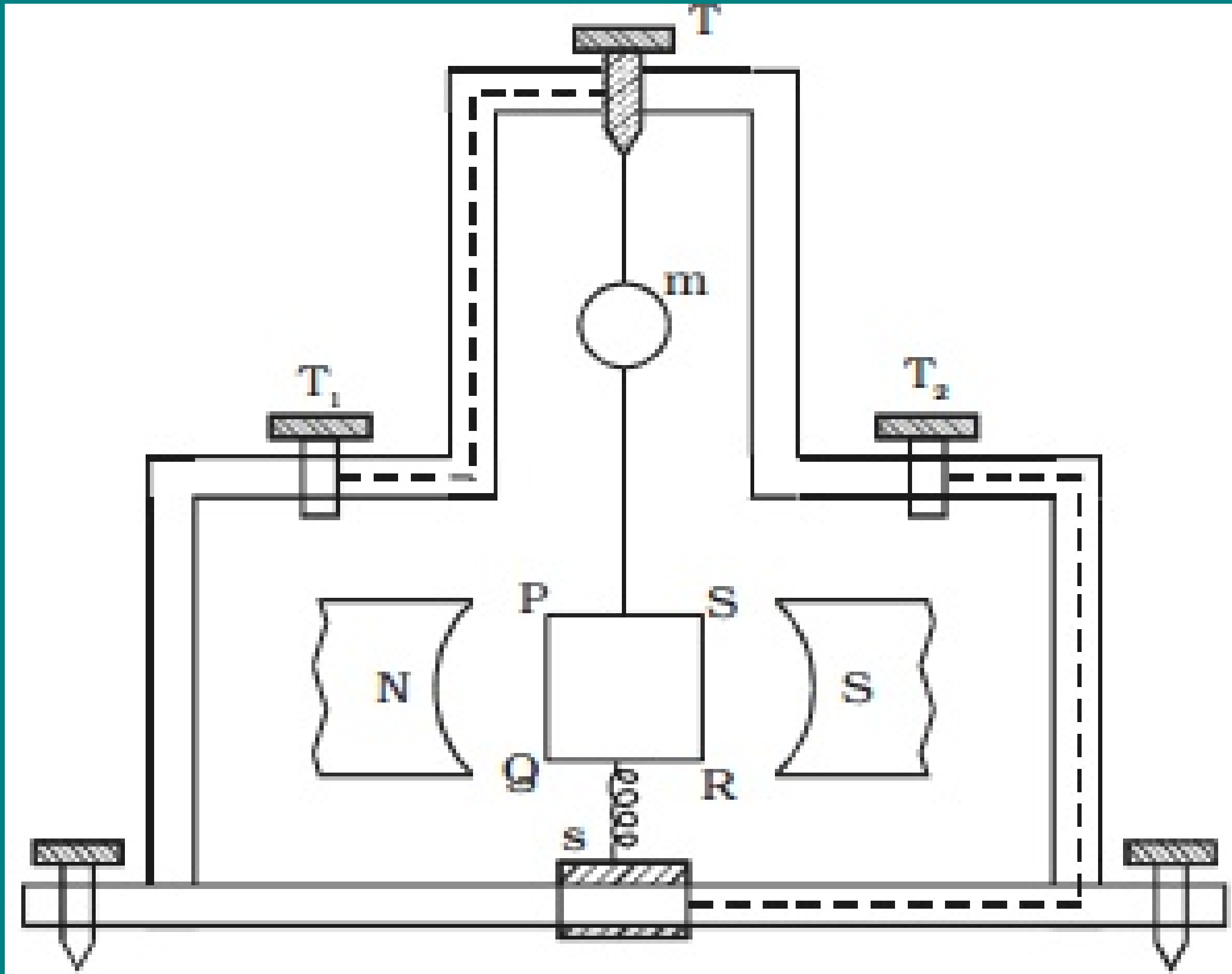
Working of A (MCG)

- In equilibrium

$$k\phi = NIAB$$
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- where, k is the **torsion constant of the spring**.

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Working of A (MCG)

- In equilibrium

$$k\phi = NIAB$$
$$\phi = \frac{NIAB}{k}$$

- where, k is the **torsion constant of the spring.**
- i.e $I \propto \phi$. Since the **deflection is directly proportional to the current** flowing through the coil, the **scale is linear** and is **calibrated to give directly the value of the current.**

THE MOVING COIL GALVANOMETER

Conversion of Galvanometer into an ammeter

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Conversion of Galvanometer into an ammeter

- The **galvanometer cannot as such be used as an ammeter to measure** the value of the **current** in a given circuit. This is **for two reasons**:

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Conversion of Galvanometer into an ammeter

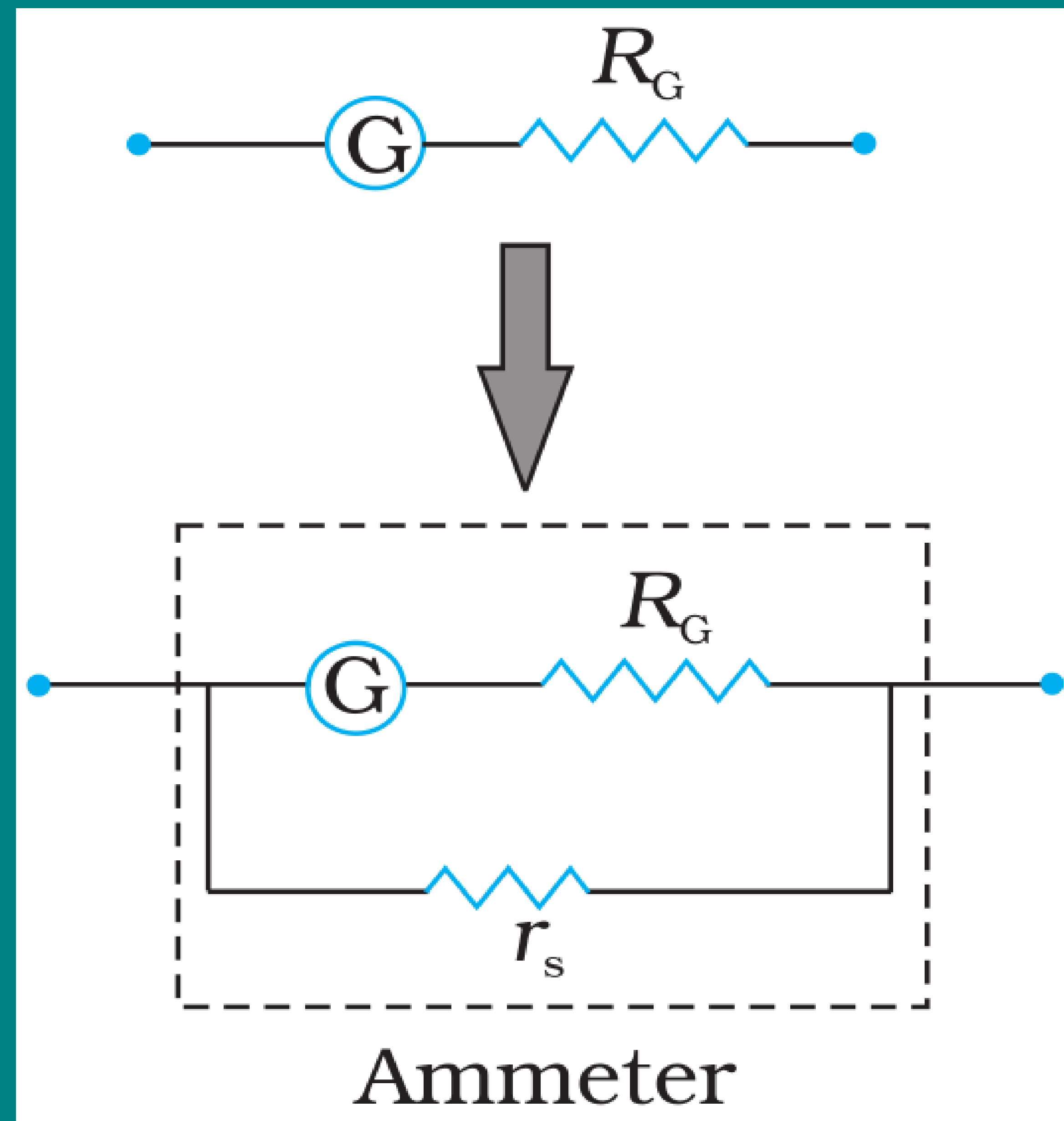
- The **galvanometer cannot as such be used as an ammeter to measure** the value of the **current** in a given circuit. This is **for two reasons**:
- **(i) Galvanometer is a very sensitive device, it gives a full-scale deflection for very small currents of the order of μA . A large current cannot be passed through the galvanometer, as it may damage the coil.**

THE MOVING COIL GALVANOMETER

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- **(ii) For measuring currents, the galvanometer has to be connected in series, and as it has a large resistance, this will change the value of the current in the circuit.**

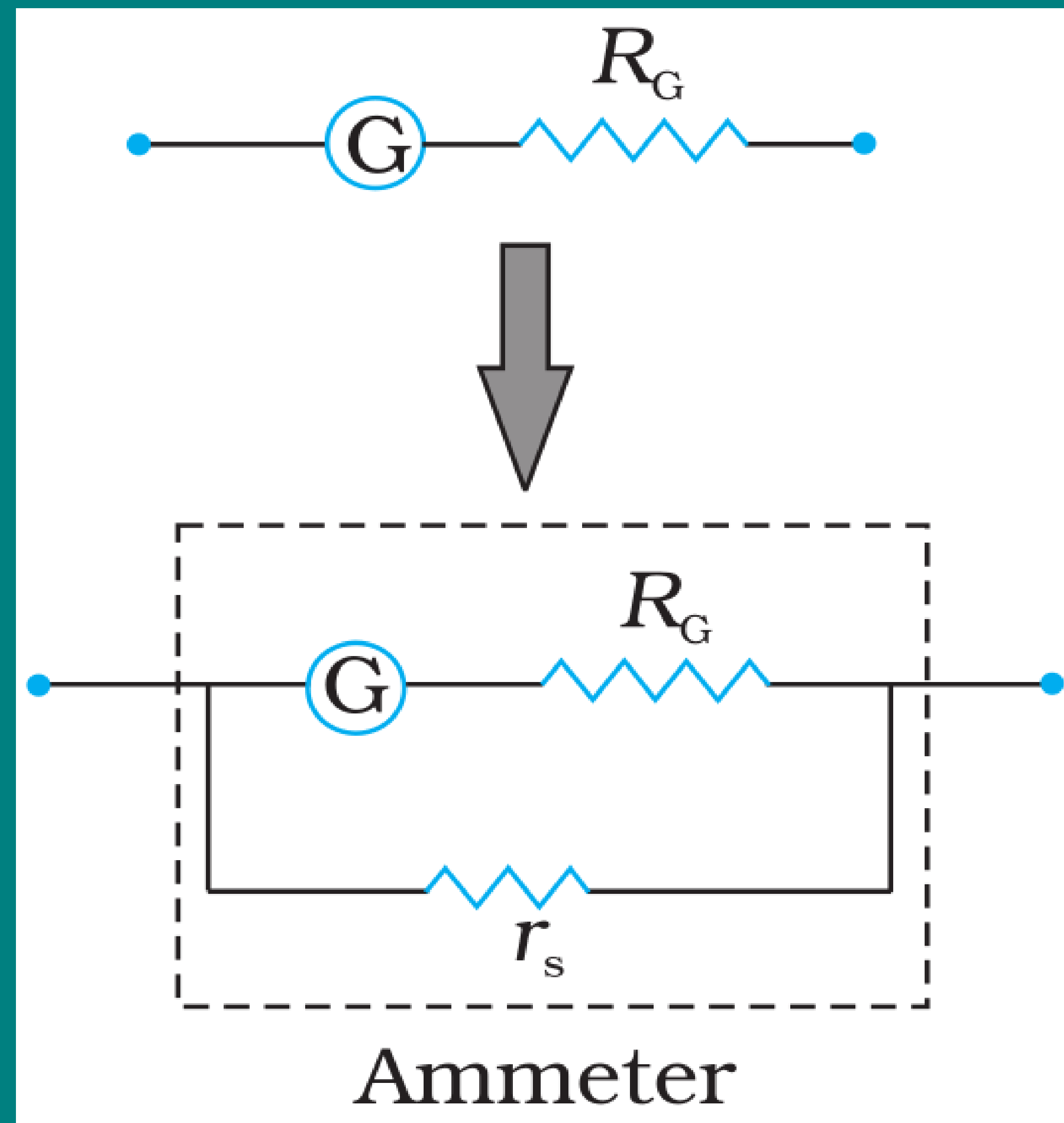
THE MOVING COIL GALVANOMETER



Conversion of Galvanometer into an ammeter

- However, a **galvanometer is converted into an ammeter by connecting a small resistance (r_s), called shunt resistance, in parallel with galvanometer.**

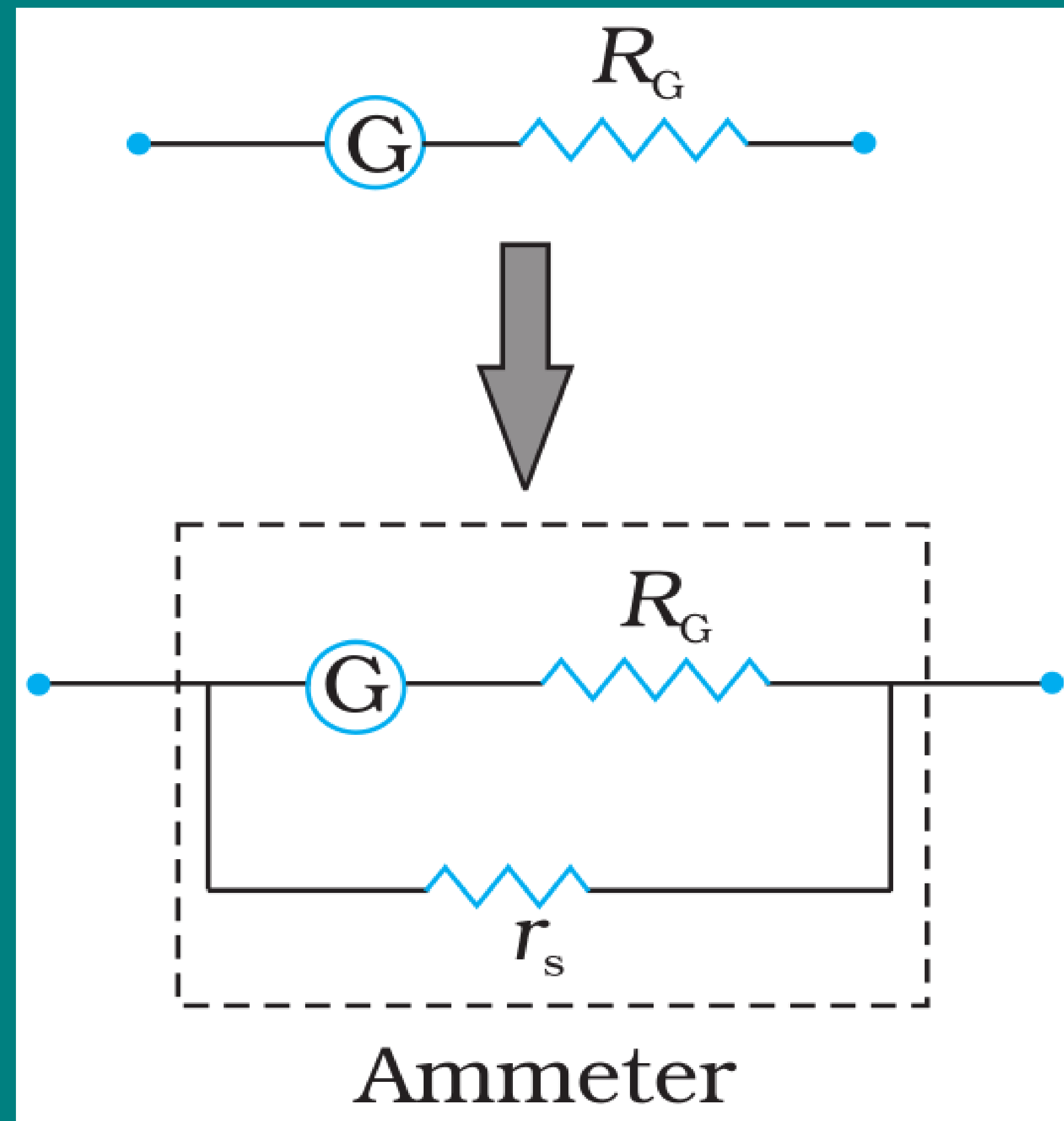
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Conversion of Galvanometer into an ammeter

- However, a **galvanometer is converted into an ammeter by connecting a small resistance (r_s), called shunt resistance, in parallel with galvanometer.**
- As a result, **when large current flows in a circuit, only a small fraction of the current passes through the galvanometer and the remaining larger portion of the current passes through the small shunt resistance.**

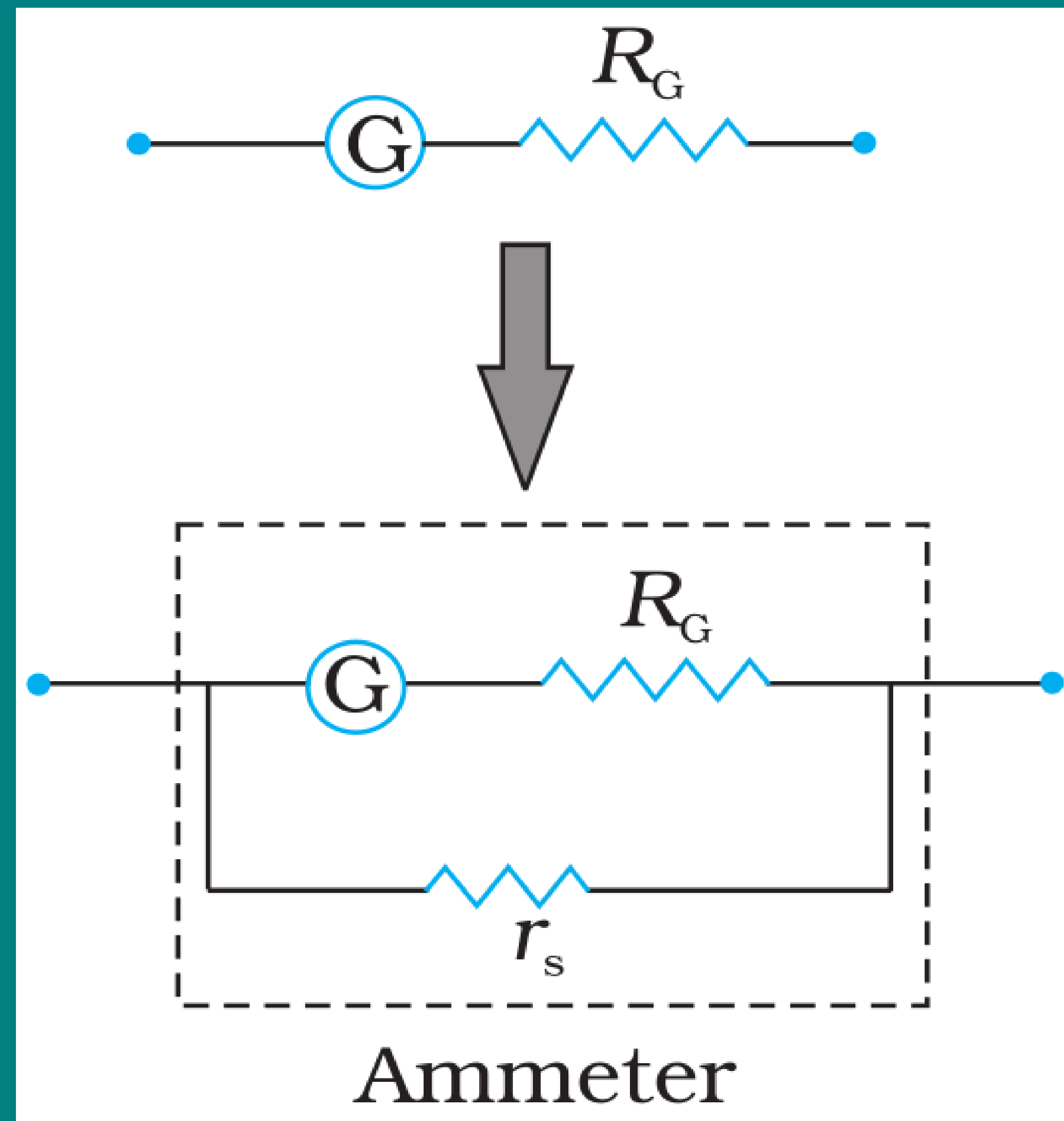
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Conversion of Galvanometer into an ammeter

- The **effective resistance of the ammeter** R_a is

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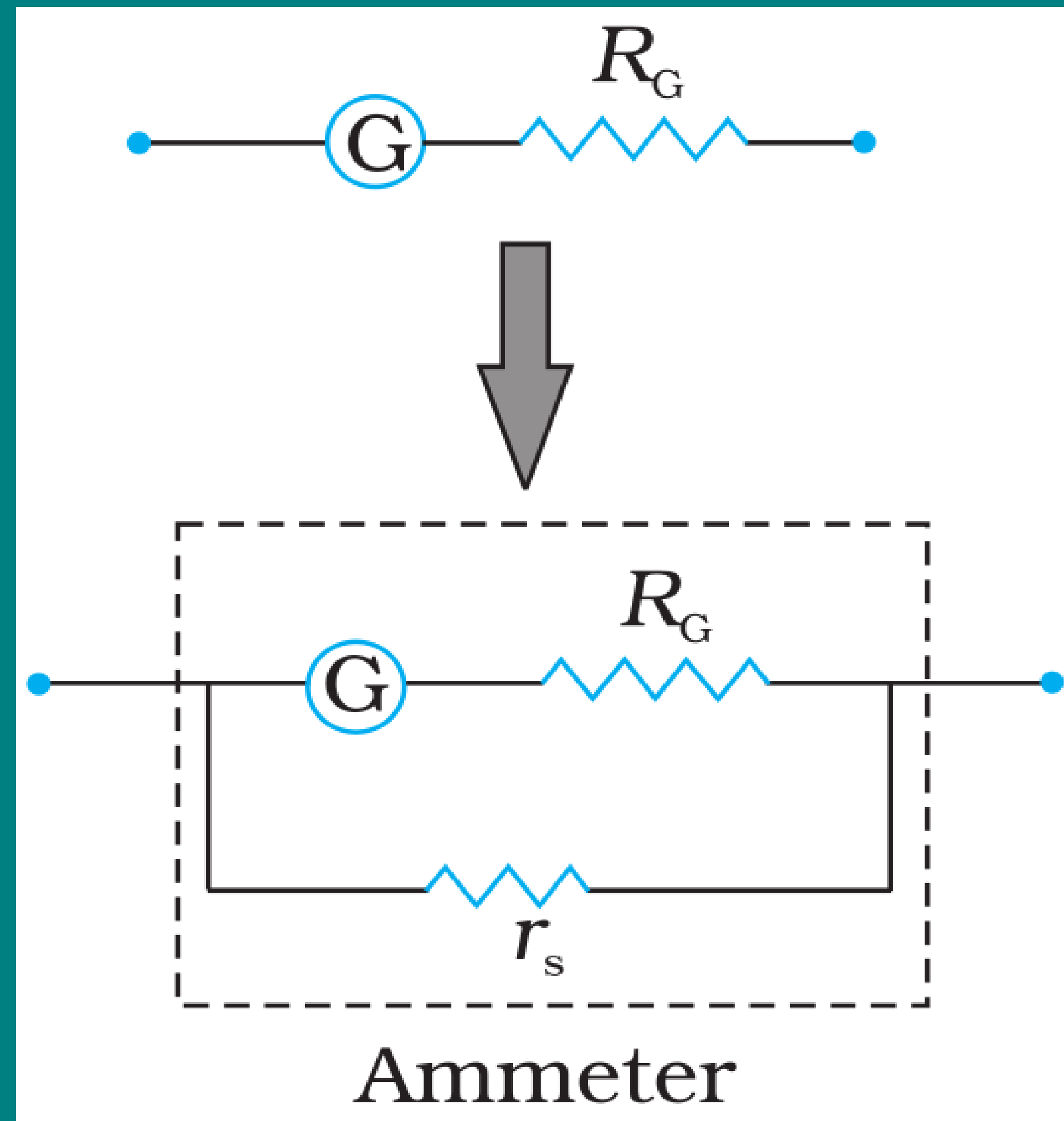


Conversion of Galvanometer into an ammeter

- The **effective resistance** of the **ammeter** R_a is

$$\frac{1}{R_a} = \frac{1}{R_G} + \frac{1}{r_s} = \frac{r_s + R_G}{r_s R_G}$$

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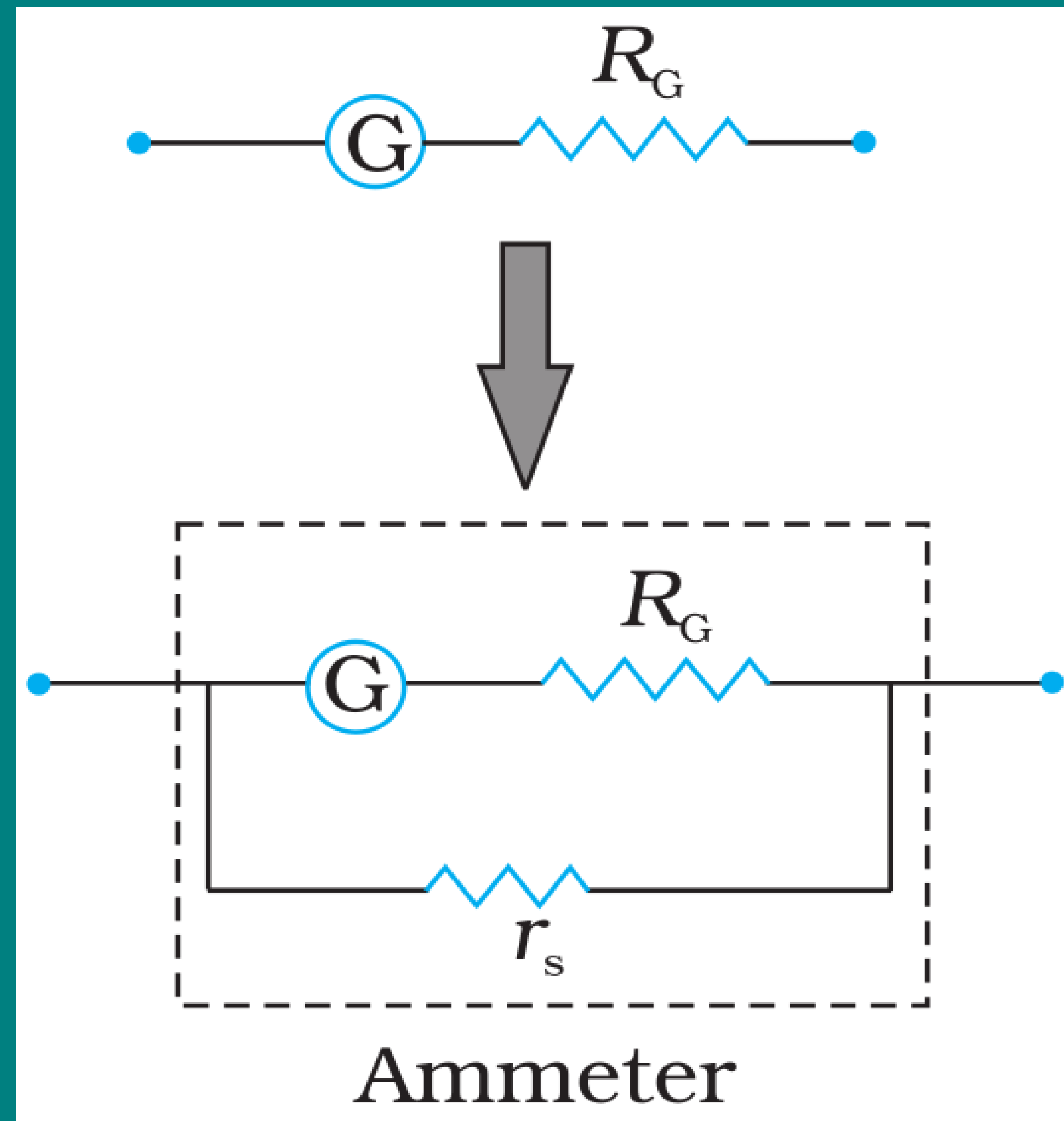
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$$R_a = \frac{r_s R_G}{r_s + R_G} \approx r_s \quad (\text{if } R_G \gg r_s)$$

THE MOVING COIL GALVANOMETER



Conversion of Galvanometer into an ammeter

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- If r_s has **small** value, the **effect of introducing the measuring instrument is also small and negligible.**

THE MOVING COIL GALVANOMETER

Current sensitivity of a Galvanometer

- The **current sensitivity** of the **galvanometer** is **defined** as the **deflection** (ϕ) **per unit current** (I).

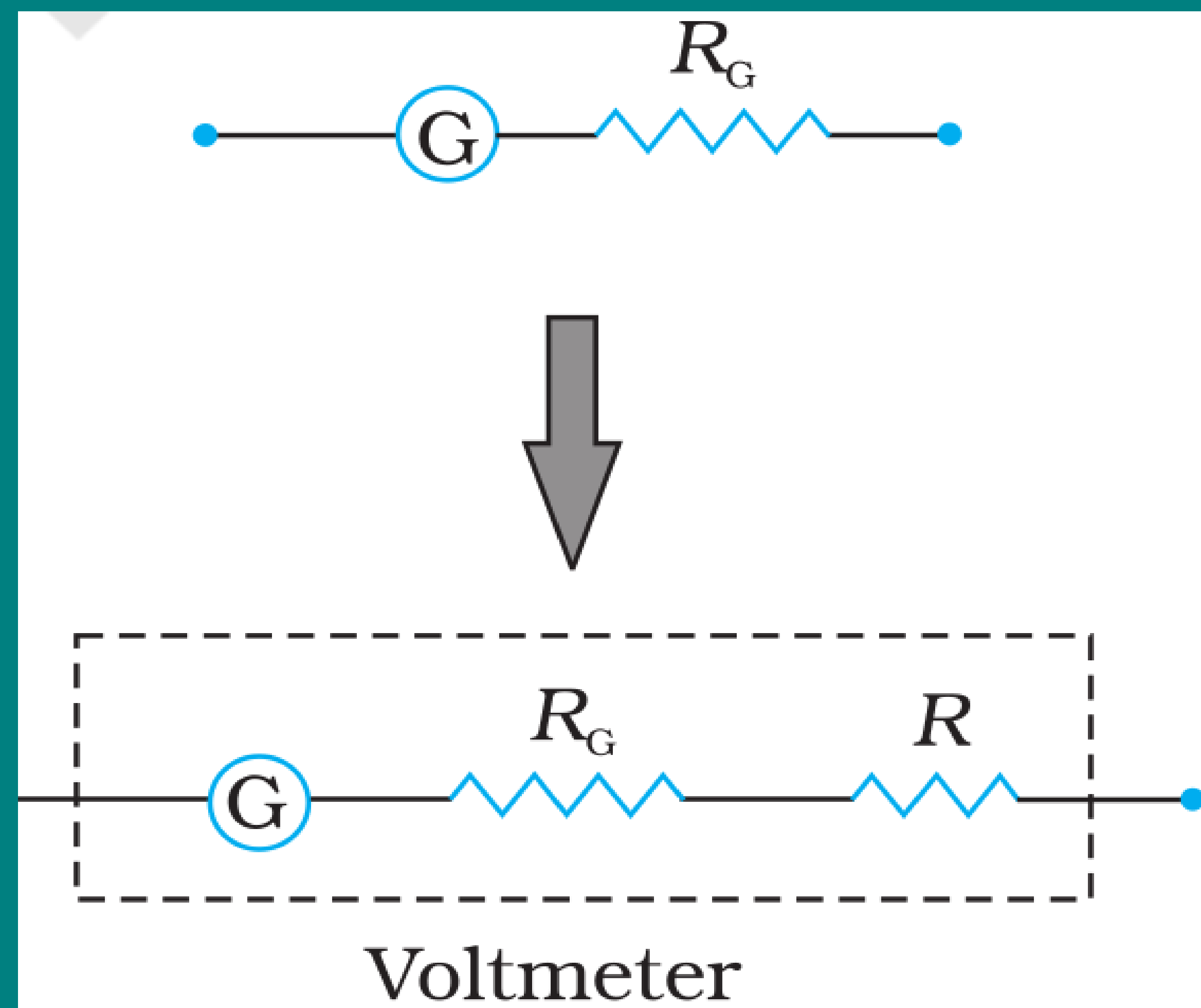
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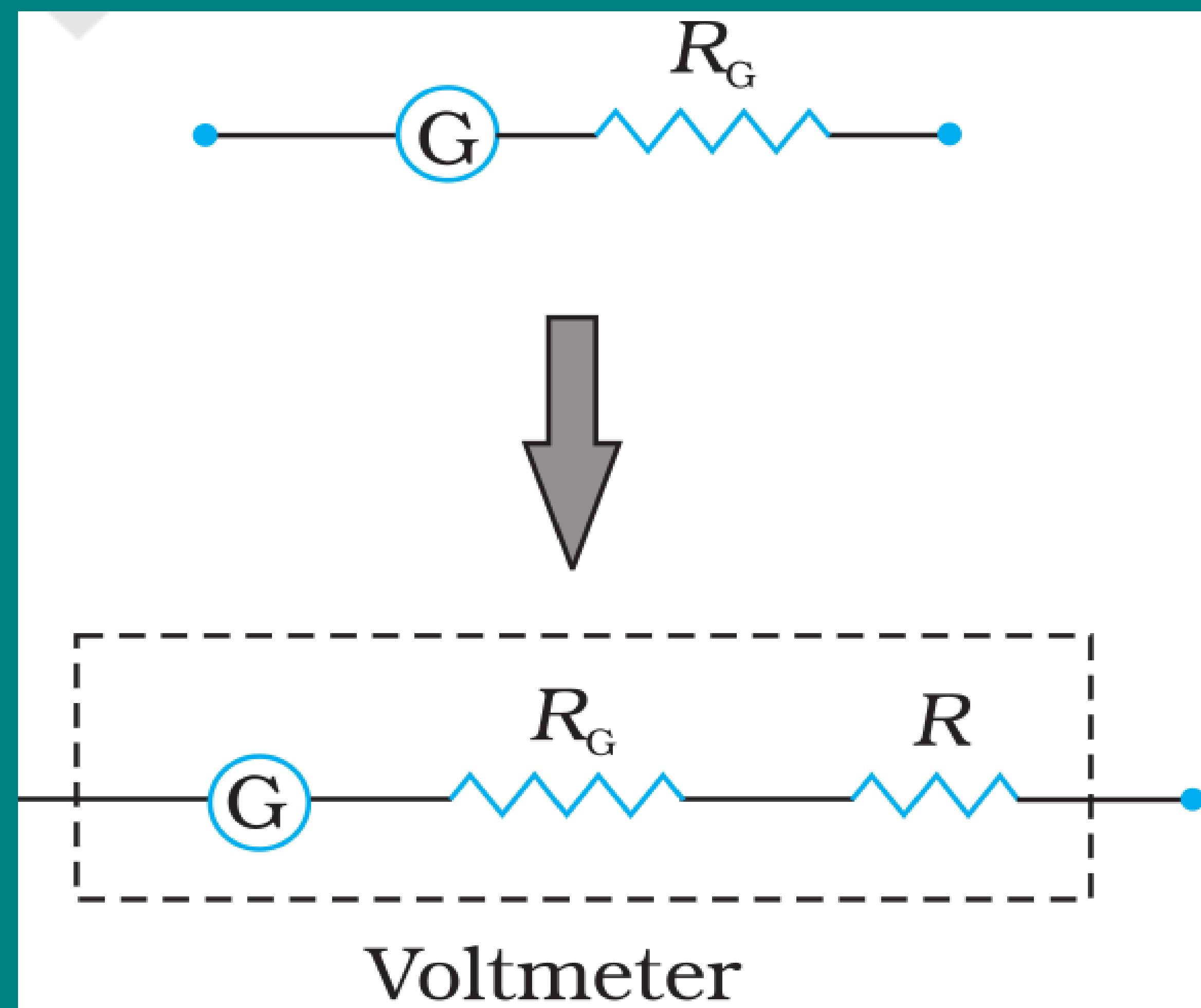
$$\frac{\phi}{I} = \frac{NAB}{k}$$

THE MOVING COIL GALVANOMETER



Conversion of Galvanometer into an voltmeter

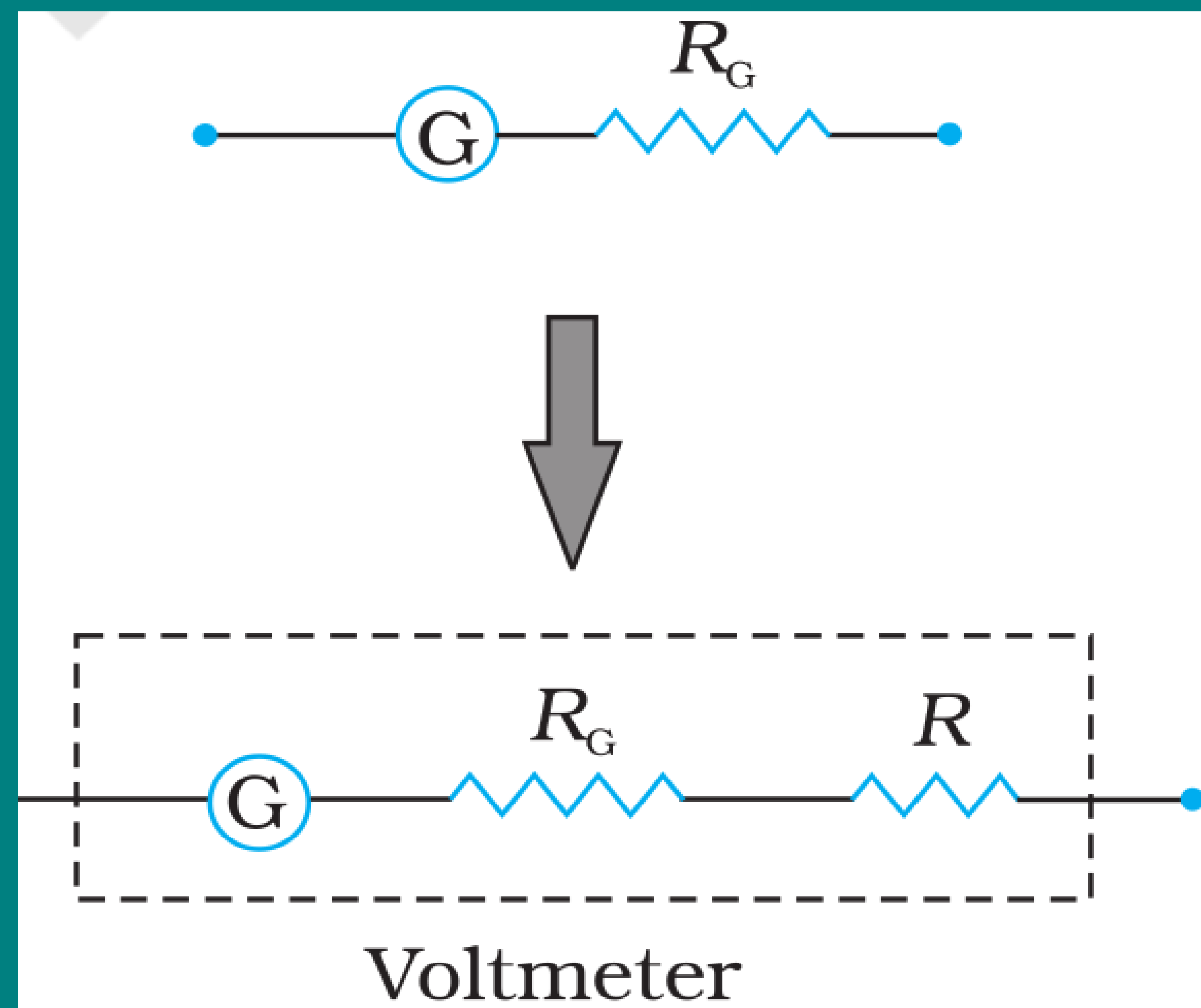
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Conversion of Galvanometer into an voltmeter

- **Voltmeter** is an **instrument used to measure potential difference** between the two ends of a current carrying conductor. it must be **connected in parallel**.

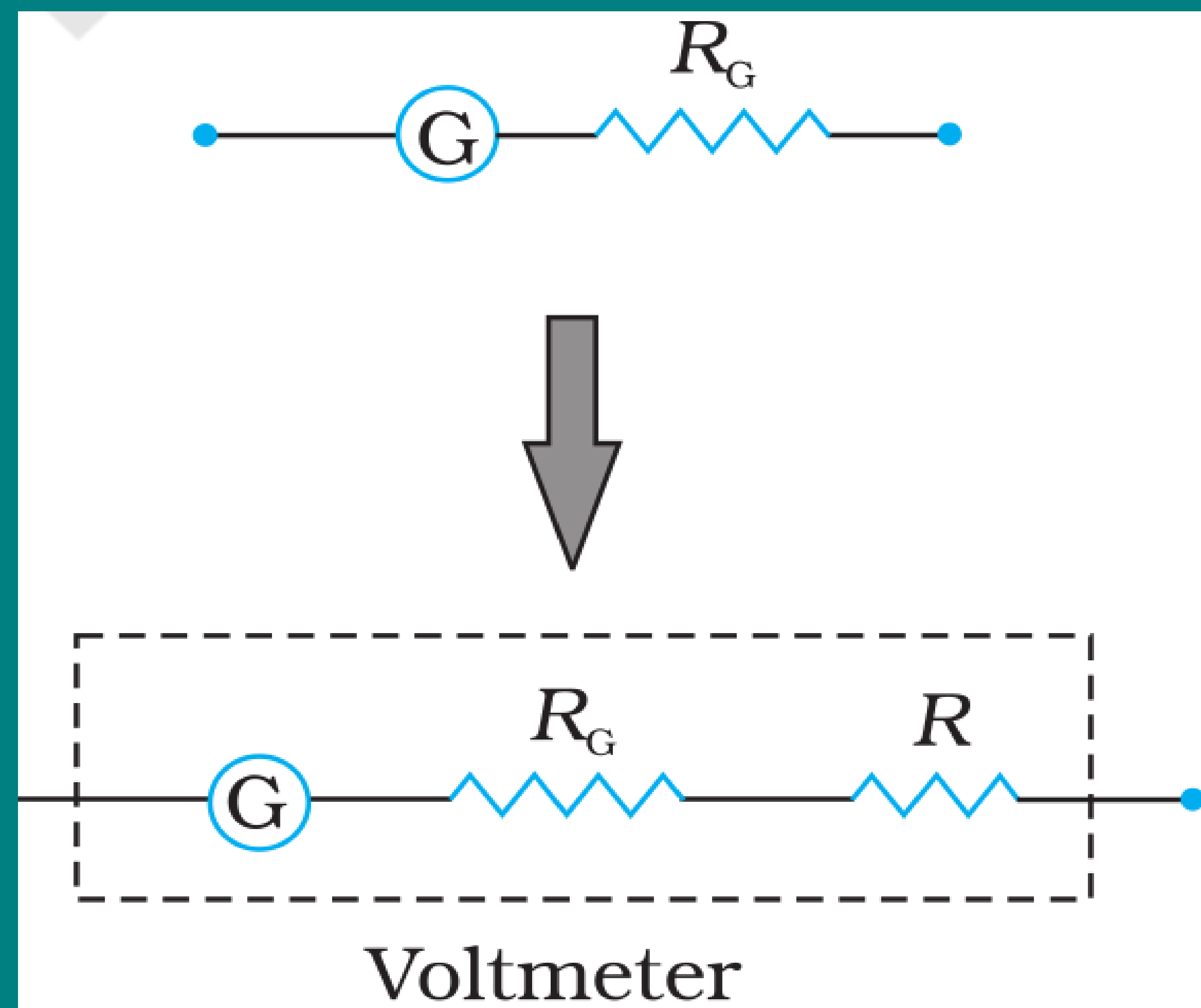
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Conversion of Galvanometer into an voltmeter

- **Voltmeter** is an **instrument used to measure potential difference** between the two ends of a current carrying conductor. it must be **connected in parallel**.
- A **galvanometer** can be **converted into a voltmeter** by **connecting a high resistance (R) in series** with it.

THE MOVING COIL GALVANOMETER



Conversion of Galvanometer into an voltmeter

- **Voltmeter** is an **instrument used to measure potential difference** between the two ends of a current carrying conductor. it must be **connected in parallel**.
- A **galvanometer** can be **converted into a voltmeter** by **connecting a high resistance (R) in series** with it.
- **Resistance of the voltmeter** is

$$R_v = R_G + R \approx R \text{ (Very large)}$$

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Conversion of Galvanometer into an voltmeter

- R_v is **very large**, and hence a **voltmeter is connected in parallel** in a circuit as it **draws the least current from the circuit.** and the **measurement of potential difference is more accurate.**

THE MOVING COIL GALVANOMETER

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- R_v is **very large**, and hence a **voltmeter is connected in parallel** in a circuit as it **draws the least current from the circuit.** and the **measurement of potential difference is more accurate.**

Voltage sensitivity of a Galvanometer

- The **voltage sensitivity of the galvanometer is defined as the deflection (ϕ) per unit voltage (V).**

THE MOVING COIL GALVANOMETER

Conversion of Galvanometer into an voltmeter

- R_v is **very large**, and hence a **voltmeter is connected in parallel** in a circuit as it **draws the least current from the circuit**. and the **measurement of potential difference is more accurate**.

Voltage sensitivity of a Galvanometer

- The **voltage sensitivity** of the **galvanometer** is **defined** as the **deflection (ϕ) per unit voltage (V)**.

$$\frac{\phi}{V} = \frac{NAB I}{kV} = \frac{NAB}{R} \left(\because \frac{I}{V} = \frac{1}{R} \right)$$

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Voltage sensitivity of a Galvanometer

- **Note: Increasing the current sensitivity may not necessarily increase the voltage sensitivity.**

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- **Note:** Increasing the **current sensitivity** may not necessarily increase the **voltage sensitivity**.
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THE MOVING COIL GALVANOMETER

Voltage sensitivity of a Galvanometer

- **Note:** Increasing the current sensitivity may not necessarily increase the voltage sensitivity.
- The current sensitivity is given by : $\frac{\phi}{I} = \frac{NAB}{k}$
- If we double the turns $N \rightarrow 2N$, then current sensitivity will also double
 $\frac{\phi}{I} \rightarrow 2\frac{\phi}{I}$.

THE MOVING COIL GALVANOMETER

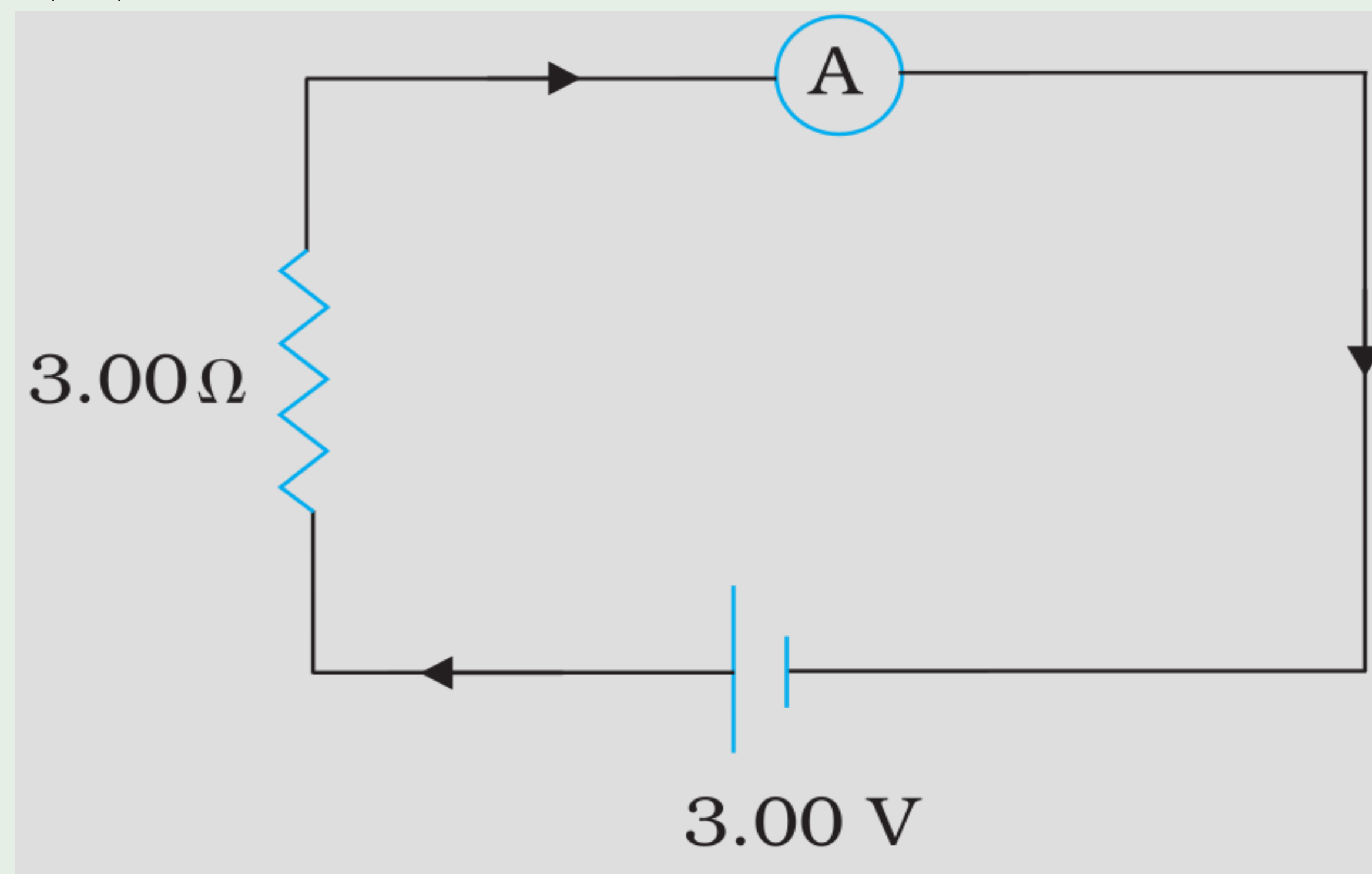
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- If we double the turns $N \rightarrow 2N$, then current sensitivity will also doubles $\frac{\phi}{I} \rightarrow 2\frac{\phi}{I}$.
- But, By doubling the turns $N \rightarrow 2N$ the resistance of the coil also doubles $R \rightarrow 2R$, thus the voltage sensitivity $\frac{\phi}{V} \rightarrow \frac{\phi}{V}$ remains unchanged.

Example 4.13

In the circuit (Fig. 4.27) the current is to be measured. What is the value of the current if the ammeter shown

- (a) is a galvanometer with a resistance $R_G = 60.00 \, \Omega$;
- (b) is a galvanometer described in (a) but converted to an ammeter by a shunt resistance $r_s = 0.02 \, \Omega$;
- (c) is an ideal ammeter with zero resistance?



Exercise 4.10

Two moving coil meters, M_1 and M_2 have the following particulars:

$$R_1 = 10 \, \Omega, \, N_1 = 30, \, A_1 = 3.6 \times 10^{-3} \, m^2, \, B_1 = 0.25 \, T$$

$$R_2 = 14 \, \Omega, \, N_2 = 42,$$

$$A_2 = 1.8 \times 10^{-3} \, m^2, \, B_2 = 0.50 \, T$$

(The spring constants are identical for the two meters).

Determine the ratio of (a) current sensitivity and (b) voltage sensitivity of M_2 and M_1 .